

Tobramycin Population Pharmacokinetics: Exploratory Data Analysis Report

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Interpretation of Results

Key Findings

Finding	Value	Assessment
Study population	300 subjects	Adequate for population PK modelling
Age	55.5 ± 14.0 yr (range 18–85)	Broad adult range; age-CLCR relationship anticipated
Body weight	76.2 ± 15.7 kg (range 45–122)	Wide range; supports weight-based covariate models
Creatinine clearance (CLCR)	97.3 ± 26.3 mL/min (range 35–130)	Variable renal function; primary source of CL variability
Sex distribution	55.7% male / 44.3% female	Balanced; sex evaluable as covariate on CL and V
Diabetes mellitus	33.0% (n = 99)	Key nephrotoxicity risk factor; evaluable as covariate
Hypertension	22.3% (n = 67)	Evaluable as secondary covariate
Prior tobramycin courses	84.3% with ≥1 prior course	High prior-exposure rate; typical of chronic infection cohort
Dosing regimen	10 doses q24h (extended-interval IV infusion)	Uniform extended-interval protocol
Weight-normalised initial dose	7.003 ± 0.042 mg/kg (CV <1%)	Protocol-defined 7 mg/kg target confirmed; negligible

Finding	Value	Assessment
		dose-BW residual
Total PK observations	7,503	Rich dataset; supports two-compartment modelling
Observations per subject	25.0 ± 1.5	Consistent sparse/intensive hybrid design
Peak concentrations (C _{max} , TSLD 0.5–2 h)	20.1 ± 5.8 mg/L (n = 490)	Within expected range for q24h extended-interval dosing
Inter-individual variability in C _{max}	28.9% CV	Substantial IIV; supports random-effects modelling on CL

Population Characteristics

The study enrolled **300 adult patients** (age 55.54 ± 14.01 years, range 18–85). The cohort was predominantly male (55.7%) and characterised by a broad body weight distribution (76.16 ± 15.69 kg, range 45–122 kg), reflecting the heterogeneous body habitus expected in a clinical infectious disease population.

Renal function, as assessed by creatinine clearance (CLCR), ranged from 35.38 to 130 mL/min (mean 97.34 ± 26.32 mL/min), spanning mildly impaired to normal function. Given that tobramycin undergoes almost exclusive renal elimination (>95% excreted unchanged in urine), this inter-individual variability in CLCR is the anticipated primary driver of clearance (CL) variability and should be the first covariate evaluated in structural model building.

Clinically relevant comorbidities were prevalent: diabetes mellitus was present in 33% of subjects and hypertension in 22.3%. Both conditions are established risk factors for aminoglycoside-related nephrotoxicity and merit evaluation as secondary covariates. Furthermore, 84.3% of subjects had received at least one prior tobramycin course, a prevalence consistent with a chronic or recurrent pulmonary infection cohort (e.g., cystic fibrosis or non-CF bronchiectasis), where multiple antibiotic courses are common.

Dosing Regimen

All subjects received tobramycin as a once-daily intravenous infusion (extended-interval dosing, q24h) for ten consecutive doses. The initial weight-normalised dose was 7.003 ± 0.042 mg/kg (coefficient of variation [CV] < 1%), confirming a **protocol-defined 7 mg/kg target dose** with negligible residual dose-weight discordance after rounding to practical dose increments. Absolute dose amounts ranged from 320 to 860 mg, directly reflecting the body weight distribution of the cohort.

This near-perfect proportionality between dose and weight has an important implication for covariate modelling: weight explains essentially all observed dose variability. Consequently, weight should be evaluated as a covariate on volume of distribution (V) using standard allometric scaling, and analysts should be mindful of the collinearity between weight and administered dose when interpreting covariate effects on CL.

PK Sampling and Observed Concentrations

A total of **7503 tobramycin plasma concentrations** were collected across 300 subjects (25 ± 1.5 samples per subject), providing a rich and consistently sampled dataset well-suited for two-compartment nonlinear mixed-effects modelling.

Peak concentrations, defined as samples collected between 0.5 and 2 hours post-dose (time since last dose, TSLD), were available for 490 observations. The observed mean C_{max} was **20.08 ± 5.81 mg/L** (CV 28.9%), representing substantial inter-individual variability. For once-daily extended-interval dosing at 7 mg/kg, clinical targets typically specify C_{max} values of 25–30 mg/L for certain indications. The observed mean C_{max} of approximately 20 mg/L may partially reflect variability in the time of the peak sample within the 0.5–2 h window rather than a systematic under-exposure; confirmation requires examination of individual TSLD-concentration profiles.

The individual TSLD-concentration profiles demonstrated consistent multi-exponential decline from peak to trough within each dosing interval, with drug concentrations approaching the limit of detection by 24 h in most subjects — a pharmacokinetic pattern consistent with the two-compartment behaviour of aminoglycosides.

Recommendations

1. **Prioritise CLCR as the primary covariate on CL.** The wide range of CLCR values (35–130 mL/min) and the known renal elimination pathway of tobramycin provide strong pharmacological justification. A power-function model ($CL = TV_CL \times (CLCR/CLCR_ref)^\theta$) is the recommended starting form for the covariate relationship.
2. **Evaluate weight using allometric scaling on V.** Given the broad weight range (45–122 kg) and protocol-defined 7 mg/kg dosing, allometric exponents of 1 on CL and 1 on V are physiologically justifiable prior values.
3. **Assess sex, diabetes, and hypertension as secondary covariates.** These variables are complete, static, and clinically plausible modulators of renal function and, consequently, tobramycin clearance. They should be evaluated after the primary structural covariate (CLCR) is established.
4. **Investigate the influence of prior tobramycin exposure (n_prior_courses).** The high proportion of previously treated subjects (84.3%) and the categorical distribution of prior course counts justify its evaluation, particularly in relation to any progressive renal adaptation or nephrotoxicity.
5. **Confirm peak sampling times before C_{max} target assessment.** The mean observed C_{max} (20.1 mg/L) is below the typical 25–30 mg/L extended-interval target. Graphical review of individual TSLD distributions within the 0.5–2 h window

should be performed to determine whether this reflects true under-exposure or sampling variability.

6. **Proceed to two-compartment structural model development.** The distribution phase visible in the individual TSLD-concentration plots, combined with the sampling density (25 observations per subject), supports estimation of a two-compartment model with inter-individual variability on CL, V1, Q, and V2.

Demographics

Variable	N	Mean (SD)	Median	Range	n (%)
Overall N	300				
Age (years)	300	55.54 (14.01)	54.92	18.00 – 85.00	
Body weight (kg)	300	76.16 (15.69)	77.77	45.00 – 122.15	
CLCR (mL/min)	300	97.34 (26.32)	99.72	35.38 – 130.00	
Baseline SCr (mg/dL)	300	0.88 (0.20)	0.85	0.44 – 1.94	
Sex, n (%)	300				
Male	167				167 (55.7%)
Female	133				133 (44.3%)
Diabetes, n (%)	300				
No	201				201 (67.0%)
Yes	99				99 (33.0%)
Hypertension, n (%)	300				
No	233				233 (77.7%)
Yes	67				67 (22.3%)
No. of prior tobramycin courses, n (%)	300				
0 courses	47				47 (15.7%)
1 course	76				76 (25.3%)
2 courses	88				88 (29.3%)
3 courses	55				55 (18.3%)
4 courses	24				24 (8.0%)
5 courses	7				7 (2.3%)
6 courses	3				3 (1.0%)

Continuous variables: Mean (SD), Median, and Range reported. Categorical variables: n (%) per category.

PK Sampling Scheme

Variable	Value
Total no. of tobramycin courses	300
Total no. of observed tobramycin concentrations per subject, mean (SD)	25.0 (1.5)
Total no. of observed tobramycin concentrations	7503
Initial tobramycin dose (mg), mean (SD)	533.3 (109.9)
Initial tobramycin dosage (mg/kg), mean (SD)	7.003 (0.042)
Total no. of peak concentrations (0.5–2 h postdose, TSLD)	490
Tobramycin peak concentration value (0.5–2 h postdose, TSLD) (mg/L), mean (SD)	20.08 (5.81)

TSLD = time since last dose. Peak concentrations defined as observations collected 0.5–2 h post-dose.

Visualizations

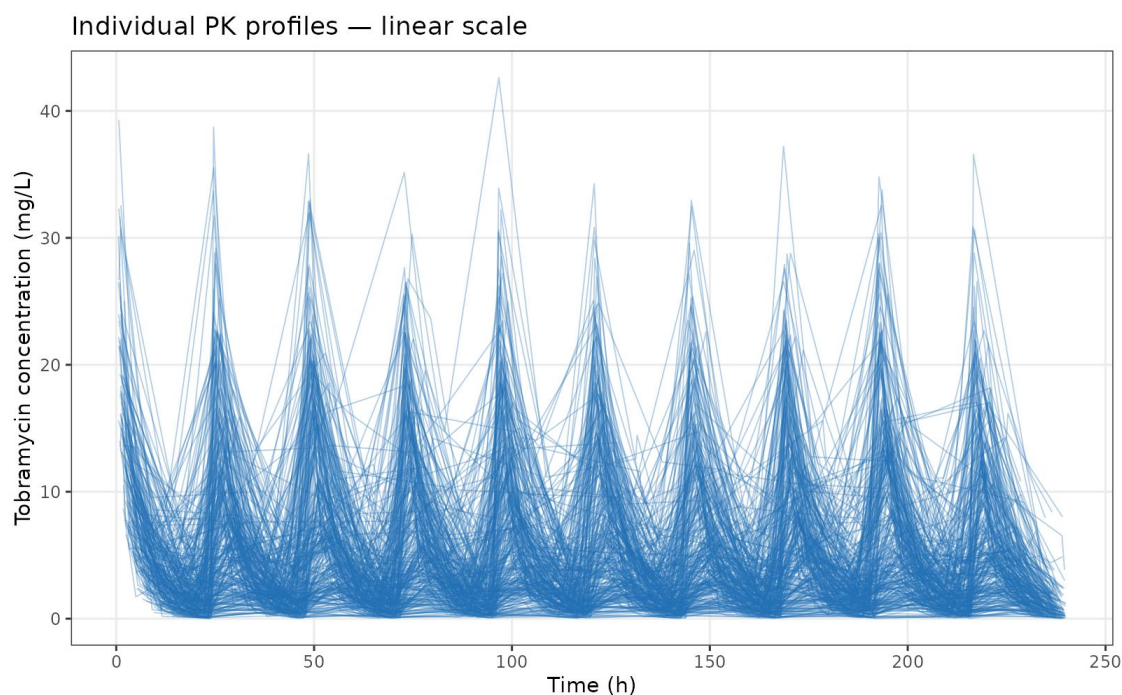


Figure 1: Individual tobramycin plasma concentration–time profiles (linear scale). Each line represents one subject. Grey spaghetti overlay illustrates the range of observed PK behaviour across 300 subjects receiving q24h extended-interval dosing.

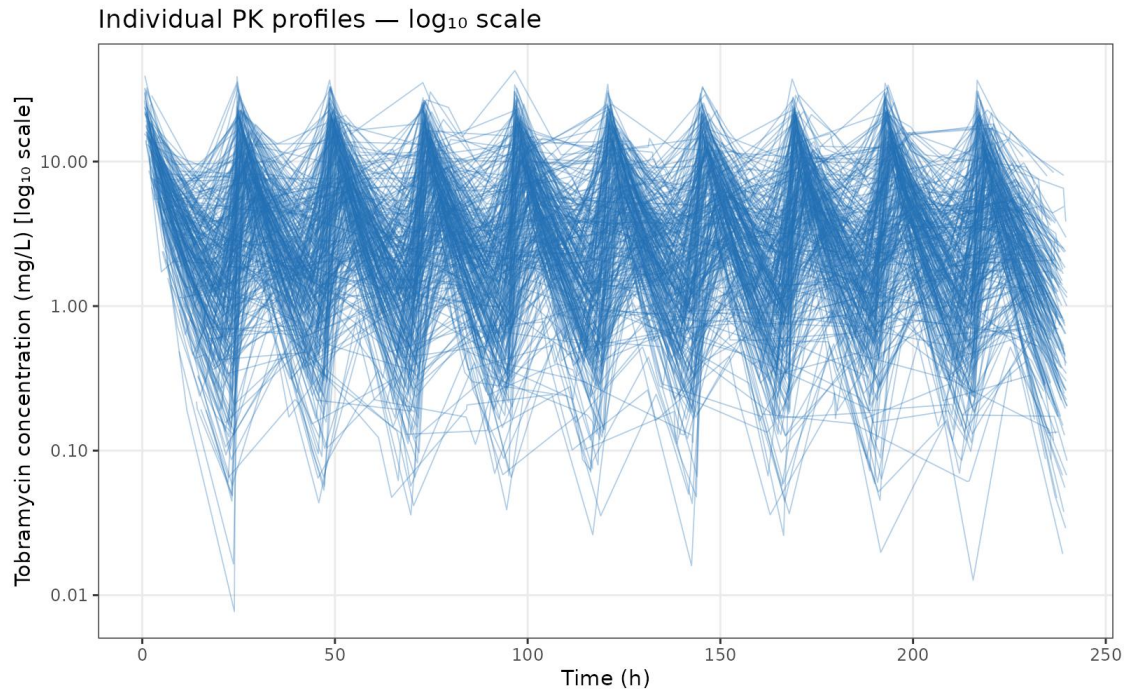


Figure 2: Figure 2. Individual tobramycin plasma concentration–time profiles ($\log_{10}$ scale). The log-linear representation reveals multi-exponential decline between doses and confirms the absence of zero or implausible concentration values.

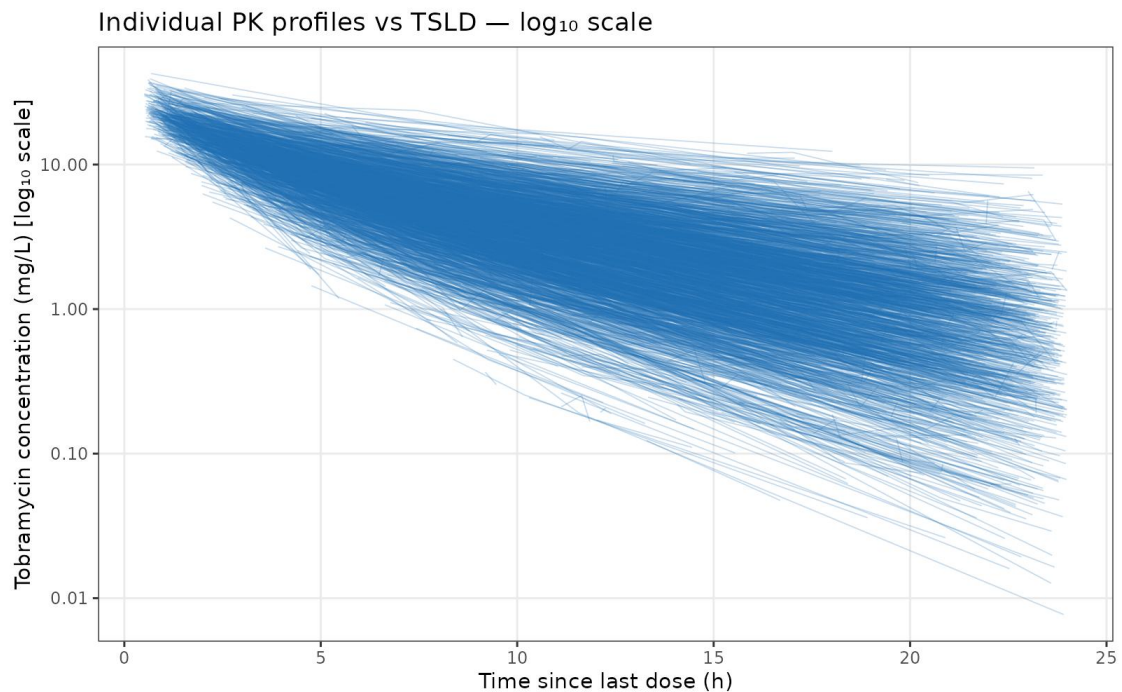


Figure 3: Figure 3. Individual tobramycin concentration profiles overlaid by dose occasion versus time since last dose (TSLD, $\log_{10}$ scale). Alignment by TSLD removes the time offset between dose events and allows direct visual comparison of within-interval PK behaviour across all dosing occasions. The consistent profile shape indicates limited between-

occasion variability.

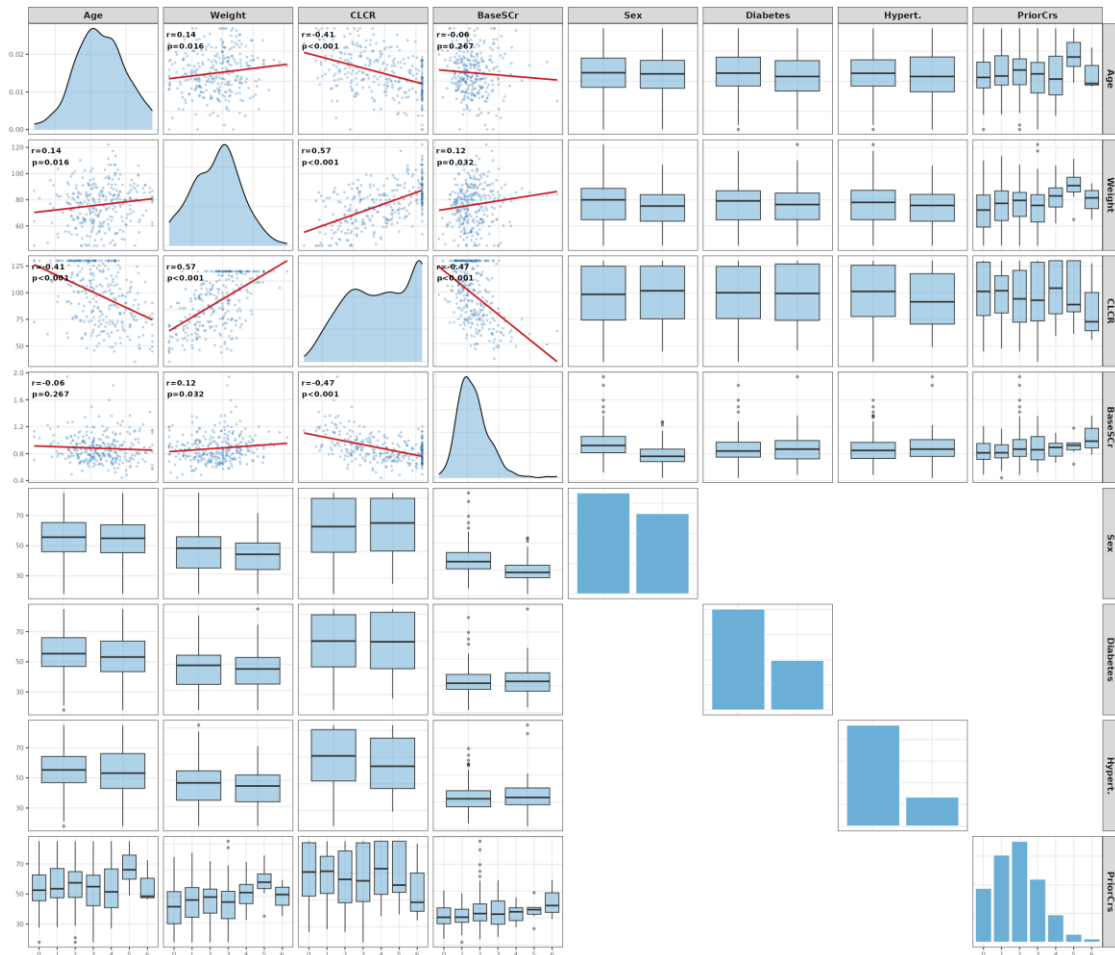


Figure 4: Figure 4. Pairwise covariate correlation matrix. Lower and upper triangle panels for continuous variable pairs show individual data points, ordinary least-squares regression lines, and annotated Pearson correlation coefficients with associated p -values. Mixed panels (one continuous, one categorical variable) display box-and-whisker plots with the categorical variable on the x-axis. Diagonal panels show marginal distributions (density for continuous variables; frequency bars for categorical variables).